Kawasaki Disease

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Introduction

- ☑ Kawasaki disease (KD) is a medium vessel vasculitis with predilection for the coronary arteries that presents as an acute, self-limited febrile illness.
- ☑ It predominantly affects, but not exclusively, children <5 years of age.
- ☑ KD is now the most common cause of acquired heart disease in children.
- ☑ Diagnosis rests on the recognition of a temporal sequence of characteristic constellation of clinical findings and the exclusion of other clinically similar entities with known causes.
- ☑ There is no pathognomonic laboratory test for diagnosis of KD.

Epidemiology

- ☑ The cause of KD still remains unknown.
- ☑ Incidence of KD is increasing; even in India.
- ☑ Asian children, especially of Japanese ancestry, have the highest relative risk for development of KD.
- ☑ May be triggered by environmental or infectious agent (viral and/or bacterial).
- ☑ Probable genetic predisposition; polygenic with genes identified that influence risk of KD and coronary artery involvement.

Diagnosis

Diagnosis of KD is based on the American Heart Association Criteria.

Classic KD is diagnosed in the presence of fever for at least 5 days together with at least four out of five principal clinical features in absence of other known diseases:

- ☑ Fever for at least 5 days
- ☑ Principal clinical features (at least four of the five following):
  1. Erythema and cracking of lips, strawberry tongue, and/or erythema of oral and pharyngeal mucosa
  2. Bilateral bulbar conjunctival injection without exudate
  3. Rash-maculopapular, diffuse erythema, erythema multiforme like
  4. Erythema and edema of the hands and feet in acute phase and/or periungual desquamation in subacute phase
  5. Cervical lymphadenopathy (≥1.5 cm diameter), usually unilateral
- ☑ Illness not explained by other known diseases.

Exceptions to the Above

- ☑ In presence of fever and coronary artery involvement on Echo, <4/5 criteria sufficient. In 25% cases, presence of coronary artery aneurysms (CAAs) is thus diagnostic.
- ☑ *Incomplete KD*: Presence of fever with less than four out of the five principal clinical criteria with compatible laboratory or echocardiography findings suggest incomplete KD. Often seen in infants <6 months and children >6 years of age, the incomplete clinical picture often delays the diagnosis. Be aware of KD in young children with unexplained fever for >5 days.
- ☑ *Atypical KD*: Patients who along with the usual clinical features of KD also have few unusual clinical manifestations such as pulmonary involvement and renal impairment and are diagnosed to have atypical KD.

Other Common Clinical Manifestations

Irritability (aseptic meningitis), arthritis (at onset or delayed), sterile pyuria (urethritis), gastroenteritis (abdominal pain, vomiting, and diarrhea), Bacillus Calmette–Guérin (BCG) reactivation, uveitis, perianal and periungual desquamation, gallbladder hydrops, and jaundice.

Complications

- ☑ *Cardiac involvement*: Myocarditis, pericarditis, cardiac failure, and valvular regurgitation
- ☑ May be complicated by macrophage activation syndrome (MAS)
- ☑ KD shock syndrome characterized by hypotension or poor perfusion
- ☑ Coronary artery disease in KD.

Risk Factors

- ☑ Male sex
- ☑ Infants <1 year or >9 years of age
- ☑ Prolonged fever
- ☑ Thrombocytopenia
- ☑ Hyponatremia.

Investigations

- ☑ *Complete blood count*: Mild-to-moderate normocytic, normochromic anemia, polymorphonuclear leukocytosis, and thrombocytosis classically occur in second week of illness with return to normal by 4–8 weeks. Thrombocytopenia is uncommon but can occur in first week. Thrombocytopenia may be a marker of MAS and is a risk factor for development of CAA.
- ☑ *Acute phase reactants* like erythrocyte sedimentation rate (ESR) and C-reactive protein (CRP) are almost always elevated.
- ☑ Elevated serum transaminases (50%).
- ☑ Hypoalbuminemia and hyponatremia.
- ☑ Sterile pyuria (>10 cells/high-power field) and with sterile cultures.

Investigations

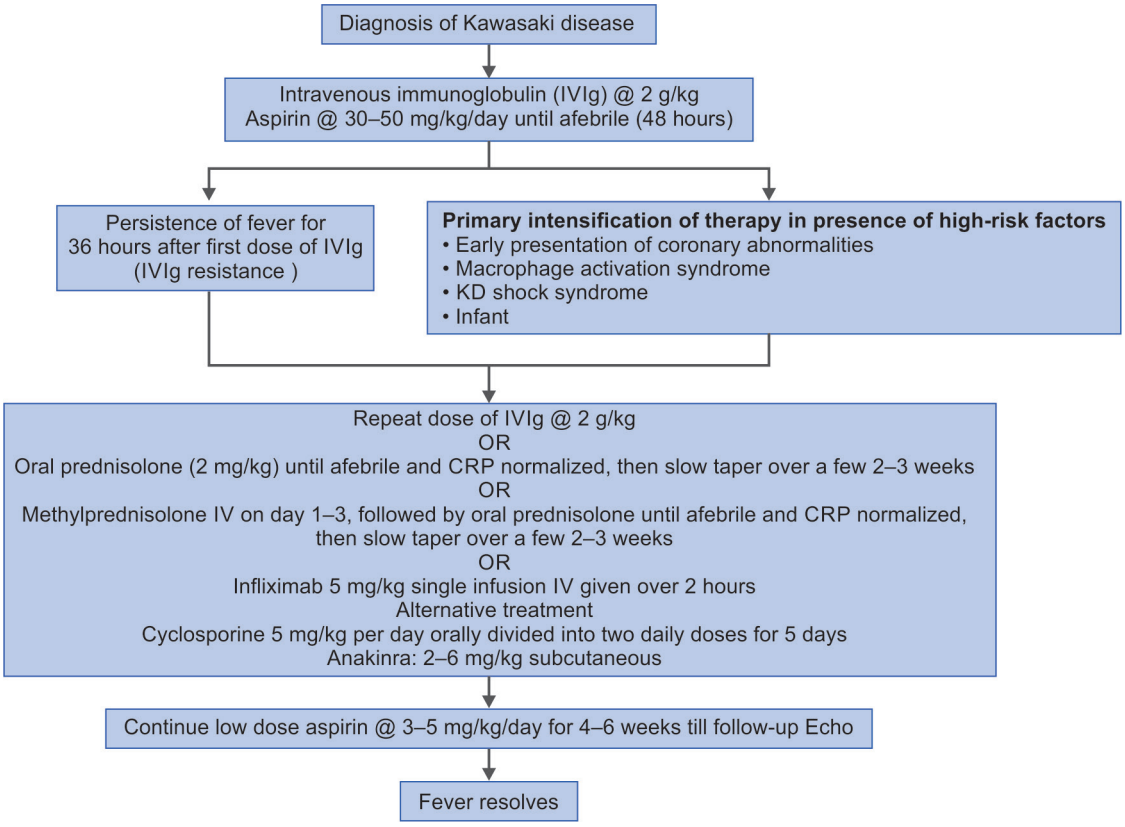
- ☑ Serum pro-brain natriuretic peptide (*proBNP*) and N-terminal proBNP (NT-proBNP) levels are elevated in KD. Serum levels of NT-proBNP > 225 pg/mL can assist in the diagnosis of KD.
- ☑ *Electrocardiogram (ECG)* may reveal evidence of myocarditis and conduction disturbances.
- ☑ *Ultrasound of the abdomen*: Hepatomegaly and/or acalculous cholecystitis (gallbladder hydrops).
- ☑ *Serum ferritin in case MAS* is suspected in patients with severe clinical disease associated with minimally elevated ESR and markedly elevated CRP.
- ☑ Serial echocardiograms are required at the time of diagnosis and 6 weeks later.

Echocardiography

- ☑ Needed for diagnosis, risk stratification, treatment planning, prognostication, and follow-up of any suspected or confirmed KD.
- ☑ Echo must always be performed at diagnosis.
- ☑ Left main coronary artery (LMCA) bifurcating into left anterior descending artery (LAD) and circumflex (Cx), right coronary artery (origin, mid, and distal segments) should be visualized. Z scores are then calculated as per body surface area (BSA).
- ☑ The coronary involvement as per Z score classification is as follows: (1): No involvement: Z score always <2 and (2) Dilatation only: 2 to <2.5. (3) small CAA: 2.5 to <5 mm; (4) medium CAA: 5 to <10 mm and absolute dimension <8 mm; (5) large/giant CAA: >10 mm or absolute dimension >8 mm. Size is calculated according to internal lumen diameter normalized for body surface area and Z score calculated.
- ☑ Perform in uncomplicated patients at 1–2 weeks and also 4–6 weeks after treatment.
- ☑ In patients with significant and evolving coronary abnormalities, Echo at least twice per week till luminal dimensions stabilize and after that at 2 weeks, 4–6 weeks, 3 months, and then every 6–12 months till parameters normalize.

- ☑ Patients with complete and incomplete forms of KD should be treated with intravenous immunoglobulins (IVIgs) preferably *within 10 days of onset of fever*.
- ☑ IVIg (2 g/kg) unequivocally reduces the occurrence of CAAs.
- ☑ In case patient reports after 10 days, IVIg should *still be given* if there is evidence of persistent fever or coronary artery abnormalities or ongoing systemic inflammation (elevated ESR/CRP).
- ☑ It is recommended to start with high-dose aspirin (80–100 mg/kg/day) (anti-inflammatory) until afebrile, then switched to low-dose 3–5 mg/kg/day (antiplatelet). Many centers now start with low-dose aspirin (3–5 mg/kg/day).

Flowchart 1: Management of Kawasaki disease.



Intravenous Immunoglobulin-resistant Kawasaki Disease

- ☑ Defined as recrudescent or persistent fever at least 36 hours following completion of the first dose of IVIg.
- ☑ 10–15% patients with KD are refractory to IVIg.
- ☑ Prompt initiation of second-line therapy is crucial in these patients.
- ☑ Options are repeat dose of IVIg; corticosteroids; and biological agents like infliximab.
- ☑ Addition of corticosteroids is advised in myocarditis and MAS.

Management of Cardiovascular Sequelae

- ☑ Children with coronary artery lesions (CALs) must be referred for specialist care and long-term follow-up.
- ☑ The goals of long-term management are to prevent thrombosis and myocardial ischemia while maintaining cardiovascular health.
- ☑ Drugs used for myocardial protection include α-blockers (carvedilol, metoprolol, or bisoprolol) and angiotensin-converting enzyme (ACE) inhibitors or angiotensin receptor blockers (ARBs).
- ☑ Hydroxymethylglutaryl coenzyme A-reductase inhibitors (statins) may be beneficial as an adjuvant therapy in KD patients with CAL.
- ☑ *Thromboprophylaxis*: Antiplatelet drugs like aspirin are commonly used in KD. In giant aneurysm or large distal aneurysms, a dual antiplatelet treatment with aspirin and clopidogrel is preferred. Anticoagulation with warfarin to achieve a target international normalized ratio (INR) of 2–3 is used. Low-molecular-weight heparin (LMWH) is equally effective to warfarin, used in young children in whom dosing with warfarin is difficult.

Long-term Management

- ☑ Risk stratification of coronary artery abnormalities based on echocardiographic coronary artery luminal dimensions (Z scores)
- ☑ Frequent cardiology assessments and surveillance is needed
- ☑ Lifestyle counseling, addressing psychosocial issues and transitional care are important follow-up components.

Further Reading

- ☑ Burns JC. Frequently asked questions regarding treatment of Kawasaki disease. Glob Cardiol Sci Pract. 2017:30:e201730.
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- ☑ Shenoy B, Singh S, Ahmed MZ, Pal P, Balan S, Viswanathan V, et al. Indian Academy of Pediatrics Position Paper on Kawasaki Disease. Indian Pediatr. 2020;57:1040-8.
- ☑ Tremoulet AH, Best BM, Song S, Wang S, Corinaldesi E, Eichenfield JR, et al. Resistance to intravenous immunoglobulin in children with Kawasaki disease. J Pediatr. 2008;153:117-21.

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G6PD Deficiency

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